



Attack MNIST Recognition

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What Are Adversarial Attacks?

- Adversarial attacks are carefully crafted small perturbations added to input images that are almost invisible to humans
- These tiny changes can cause a model to make completely wrong predictions with high confidence
- The key idea: maximize model error while keeping the image visually unchanged
- This exposes a major weakness in deep learning systems used in real-world applications

FGSM - The Giant Leap

- FGSM is a single-step white-box attack
- Extremely fast and computationally cheap
- Computes the gradient of the loss function only once, then updates all pixels simultaneously
- The update rule pushes each pixel in the direction that maximally increases the model's error
- Visual effect: often appears as grainy or static-like noise
- Limitation: less precise and sometimes more noticeable noise

I-FGSM - Careful Climb

- Iterative FGSM improves FGSM by breaking attack into many small steps
- Instead of one big jump, it applies small perturbations repeatedly
- Includes a clipping mechanism to ensure the image stays close to the original input
- Visual effect: subtle and targeted changes
- Main advantage: higher success rate and less visible perturbations
- Limitation: slower than FGSM due to multiple computations

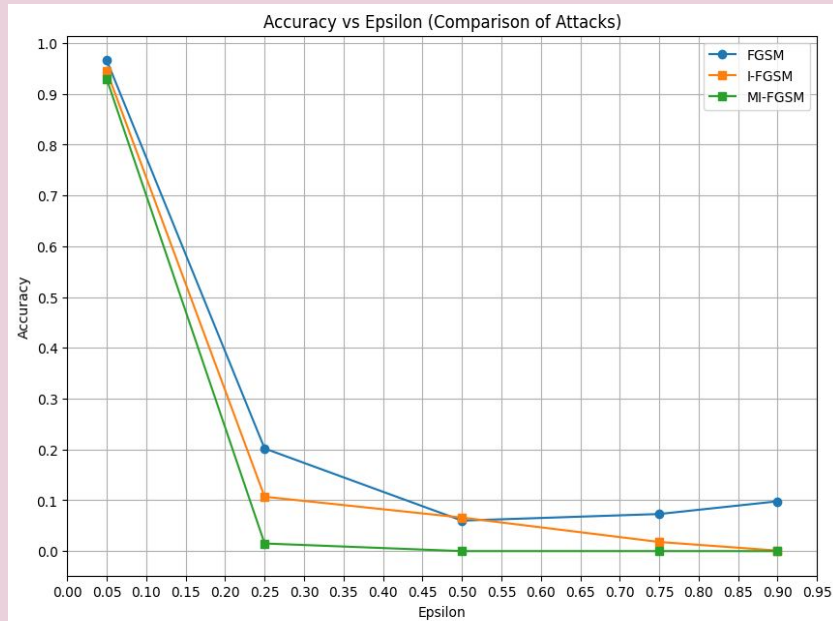
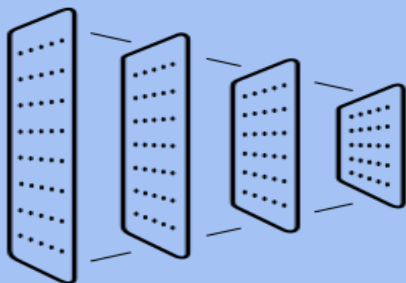
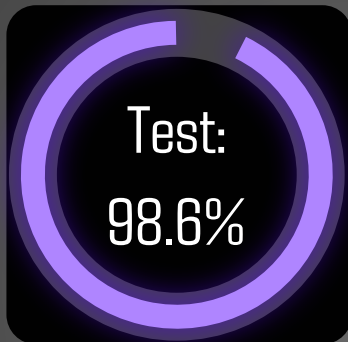
MI-FGSM

- Momentum Iterative FGSM enhances I-FGSM by adding a momentum term
- Instead of relying only on the current gradient, it accumulates past gradients to build direction
- Helps the attack avoid local maxima or unstable gradient directions
- Produces smoother perturbations that are less noisy and more structured
- Visual effect: cleaner and more natural-looking adversarial noise

Attack on CNN

Detects
patterns
like edges
and
shapes

Understands
image
structure

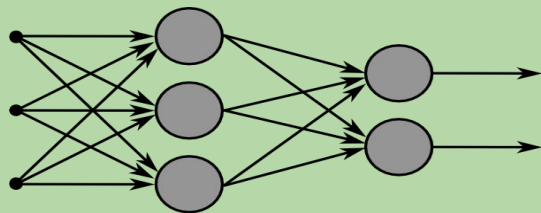


Attack on MLP

Sees
image as
just
numbers

No spatial
awareness

Test:
91.5%



hidden layer

output layer

